PSAI_016

Open Grant Proposal: Blockchain Solution for Artificial Intelligence: Machine Learning Dataset and Modeling Storage Using Filecoin

Name of Project: Blockchain Solution for Artificial Intelligence: Machine Learning Dataset and Modeling Storage Using Filecoin

Proposal Category: `app-dev`

Proposer: `RiveringAI (WillTechMusing)`

Do you agree to open source all work you do on behalf of this RFP and dual-license under MIT and APACHE2 licenses?: “Yes”

Project Description

Machine Learning (ML) has grown significantly in various industries in recent years. ML practitioners from businesses Artificial Intelligent (AI) teams to individual data scientists face multiple data storage related challenges: finding proven and good dataset, storing and maintaining them with ground truth labels, and keeping track of variances of different ML modeling files that are generated during the experiment and production lifecycle. These data files typically are stored internally, or through AWS / Azure / GCP cloud by many individuals or teams but not organized properly, difficult to version, hard to reference or share.

In particular, Deep Learning, a type of machine learning, requires data scientists to use much larger datasets to train models that imitates the way humans gain certain types of knowledge. These large datasets (for example, in NLP) can have billions of words or sentences in hundreds of GBs or TBs in size which post challenges in storage and management. Further more, these larges datasets are stored dozens or thousands of copies around the world - a big waste yet none of them are reliable.

This project aims to use Filecoin storage network as data foundation to simplify the AI/ML tasks so that scentists can focus on collect, edit, publish and share Natural Language Processing (NLP) domain open datasets, open algorithms and ML models for AI researchers and various industry applications.

- Identify, edit and manage open datasets in NLP space for machine learning and deep learning
- Leverage IPFS to store and manage list of datasets, models as well as versioning
- Filecoin is ideal solution to store these datasets and models in consistent, verifiable, decentralizing way
- Integrate with multiple Filecoin-based partners (FileSwan, Estuary, BidBot etc) on storage deals making and distribution

Value

As Artificial Intelligence (AI) continues to proliferate, more and more organizations and businesses are investing heavily into machine learning and data scientists as competitive advantages. Providing a place to share creditable open datasets, open algorithms and ML models will contribute significantly to this growing technology revolution. Levering Filecoin (and IPFS as cache) as the foundation layers will not only greatly benefit the overall Filecoin ecosystem but also contribute tremendously in preserving valuable human datasets for future AI development. - ***Drive Rapid ML Development*** - provides a list of NLP databases to all ML scientists. Open algorithms will dramatically speed up applications around these datasets - ***Reduce Cost*** - makes these datasets and algorithms shared to everyone will simplify the storage needs by individuals and teams hence reduce the overall redundant copies of these data storing hundreds of copies of them in the cloud - ***Trustable and Safe Storage*** - Filecoin storage provides trustable and safe data to prevent accidental costly damages or deletion of valuable data - ***Extend Filecoin Ecosystem with AI Applications*** - Extend Filecon ecosystem with AI / ML communities to build applications around Fileconi and IPFS to further improve values of the Filecoin in long run.

Additional Beneifts of Storaging Open Algorithms

In addtion to storing ML datasets, this tool allow users upload and store open algorithms as well. The ML algorithms (can be one or more than one) are to demonstrate how these open ML datasets are used in training and ML model development. The combination of ML datasets and algorithms can greatly simplify other users and application data scientists to rapidly use them as references to develop further or improved ML models.

Risks:

one of the risks is that this project requires Filecoin network storage to reliable in storing ML datasets, and be responsive with retrieval capability. This can be migrated by working closely with Filecoin community since the storage deals and retrieval capability are core values for the Filecoin network.

Deliverables

Please describe in details what your final deliverable for this project will be. Include a specification of the project and what functionality the software will deliver when it is finished.

- User can upload, store, and search list of creditable NLP datasets from website
- User can review the sources, references and rating of each dataset
- User can download a selected open dataset for ML training
- User can upload, review and download the open algorithms of machine learning codes that use the specific dataset as part of the training

- User can upload, and download the corresponding ML model produced by the open algorithms to run prediction

Development Roadmap

No.	Milestone Description	Funding	Estimated Timeframe
1	Project functionality design and scope definition.	NA	2 weeks
2	Project user interface design	NA	1 weeks
3	Infrastructure architecture design and implementation plan readiness.	NA	1 weeks
4	Command line tool coding	5,000	4 weeks
5	User interface coding	10,000	3 weeks
6	Dataset uploads, open ML algorithms development	10,000	4 weeks
7	Document and tutorial	TBD	1 weeks
8	Product release go marketing	5,000	1 weeks

Total Budget Requested

$30,000 for total. Please see the development roadmap for details.

Maintenance and Upgrade Plans

Will continuously take comments from the community and make improvements. There will be future enhancement and upgrade to expand more datasets, more ML algorithms, and potentially work on solutions that bring AI computing into Filecoin network.

Team

Team Members

- William Li (Main Developer Lead, github name: william8work, and Slack handle: WillTechMusing)
- Lucia Zheng (Data Scientist)
- Web Developer (To be hired)
- Charles Cao (FilSwan Support)

Team Member LinkedIn Profiles

- https://www.linkedin.com/in/williamlitech/

Relevant Experience

WillTechMusing (William Li) in an active member of MinerX and X2 program of Filecoin mining community. In addition, he is a leader in the AI community, has rich experience manage and deliver cutting edge technical projects in data analytics and AI/machine learning. He is also an active conference speaker in multiple AI and technology conferences.

AI Paper Published: - https://arxiv.org/abs/1907.03715

AI Articule Published: - https://www.destinationcrm.com/Articles/Web-Exclusives/Viewpoints/How-AI-Can-Bridge-the-Gaps-Between-CRM-Marketing-and-Sales-146140.aspx

AI Conference Speaker: - http://aisea20.xnextcon.com/